The first step of a nitration reaction involves using a mixture of sulfuric acid and nitric acid to form an NO_2^+ electrophile, as shown below.

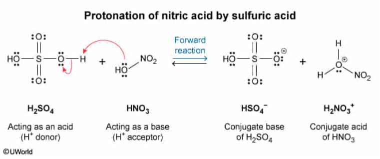
H 2 S

In the forward reaction of the equilibrium, which of the following molecules acts as a Brønsted-Lowry base?

- A. H_2SO_4 [8%]
- B. HNO_3 [74%]
- C. HSO_4^- [6%]
- D. $H_2NO_3^+$ [11%]

Correct
74% answered correctly

Explanation:



According to the Brønsted-Lowry definitions, an **acid** is a chemical species that **donates a proton** (H^+ ion), and a **base** is a chemical species that **accepts a proton**. The conjugate of an acid (or base) is the species that the acid (or base) becomes after losing (or gaining) one proton. A base is converted into its **conjugate acid** when the base accepts a proton. Likewise, an acid is converted into its **conjugate base** when the acid donates a proton. Therefore, an acid (or base) and its conjugate have chemical formulas that differ by only one proton.

Whether a molecule functions as an acid or as a base sometimes depends on the relative acidity (acid strength) of the molecule. For example, H_2SO_4 and HNO_3 both usually function as strong acids (proton donors). However, if H_2SO_4 and HNO_3 are mixed together, H_2SO_4 is demonstrated to be a stronger acid than HNO_3 because the HNO_3 is induced to *accept* another proton to form $H_2NO_3^+$.

By the Brønsted-Lowry definition, a base is a proton acceptor. Therefore, in the forward reaction of the equilibrium, HNO_3 acts as a Brønsted-Lowry base by accepting a proton from H_2SO_4.

(Choice A) H_2SO_4 functions as a Brønsted-Lowry *acid* (proton donor).

(Choice C) HSO_4^- acts as a Brønsted-Lowry base, but it does so in the *reverse* reaction of the equilibrium.

(Choice D) $H_2NO_3^+$ is the conjugate acid of HNO_3.

Educational objective:
By the Brønsted-Lowry definitions, an acid is a chemical species that donates a proton, and a base is a chemical species that accepts a proton. The conjugate of an acid (or base) is the species that the acid (or base) becomes after losing (or gaining) one proton. An acid (or base) and its conjugate have chemical formulas that differ by only one proton.

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